Water Ledger Texas: methods brief

Model version 1.2 · data snapshot 2026-10-29 · waterledger.org · github.com/northlinehydrology/waterledger

What it estimates

For any city, town, village, township or county in Texas, Water Ledger estimates how much water a data center campus of a given size and cooling design would take, use up and send back, on an average day and on a 1-in-10-year peak day, and sets that beside the public water system, the sewer plant, the streams downstream and the electric utility that would serve it. It uses public data only and is a screening estimate, not a design.

Data

Category	Source	In Texas
Places	Census TIGERweb: incorporated places; county subdivisions that are governments; counties	1,481 places: 1,227 cities, towns and villages; 254 counties
Water supply	EPA community water system service areas, with SDWIS population and connections	4,582 systems; 95% of places have one mapped
Wastewater	EPA Clean Watersheds Needs Survey 2022 (design flow) matched to ECHO (recent flow, receiving water)	1,059 plants; 91% of places have one within 40 km
Receiving waters	USGS Watershed Boundary Dataset; EPA ATTAINS (TCEQMAIN, 2026 report)	1,139 watersheds with a downstream path; 3,086 watersheds on those paths
Climate	NOAA ISD station records, 1995-2024, one station per NOAA climate division; Atlas 14 design storms; Climate at a Glance rainfall	10 climate divisions
Energy	HIFLD retail service territories (PNNL copy); EIA-861 sales and peaks; Siddik et al. 2024 water per kWh by grid region	151 utilities with loads; grid regions: EPE, ERCO, MISO, PNM, SWPP

How an estimate is made

Cooling water. Hour by hour over 30 years of weather, the campus's heat load is turned into evaporation, drift and blowdown for each cooling design: cooling towers, hybrid (evaporative only in warm weather), closed loop with evaporative assist on the hottest hours, and closed loop with no water for cooling. The evaporative share follows Rutberg et al. (2011); blowdown follows the DOE FEMP water balance.

Uncertainty. The model runs 200 times with its 15 cooling parameters drawn from published or estimated ranges (marked [A] where they stand in for a facility's own data). Results are the 10th, 50th and 90th percentiles; the peak day is the 90th percentile of each year's highest day.

Staff water and loop top-ups are added at the campus's size (people on site, and small leaks from closed loops).

Source and wastewater. The campus uses the public water system mapped in the place; with none mapped, a closed loop uses the nearest system serving over 3,300 people if it is within 10 km, and anything else uses new wells. Wastewater goes to the sewer when a plant is in reach and, for evaporative designs, has room for the blowdown; otherwise a septic system (closed loop) or the campus's own discharge.

Downstream. Each place's main watershed is followed downstream (up to 25 watersheds, across state lines), with the impairments and cleanup plans (TMDLs) EPA lists for each.

How this differs from Michigan

Weather. Texas uses NOAA station records (Lubbock International Airport; Abilene Regional Airport; Fort Worth Naval Air Station Jrb/Carswell Field; Angelina County Airport; Winkler County Airport; Kimble County Airport; Victoria Regional Airport; G Bush Intercontinental Ap/Houston Ap; Cotulla-La Salle Co Arpt; Mc Allen Miller Intl Arpt). Michigan uses ERA5 reanalysis. For cooling towers the two agree within about 1% in Michigan, but for hybrid and assist, which use water only on the hottest hours, station records give higher averages (4-50% higher in southern Michigan, 2-5 times at Traverse City and Alpena), so hybrid and assist figures are not strictly comparable across the two.

Not yet included here: state-reported water use by sector, household wells, coldwater streams, lake-source cooling and state rules ("state rules not yet reviewed"). Utilities are listed by customers served, since the national territory map draws many small utilities as whole counties.

Example: 300 MW of IT capacity in Austin

Million gallons a day: median, with the 10th to 90th percentile range in brackets. Peak day is the 1-in-10-year highest day. "Used up" is water evaporated or otherwise not returned. Defaults for water source and wastewater are filled in by the rules above (muni: public system; sewer: the nearest plant).

Design	Withdrawn, average day	Withdrawn, peak day	Used up, average day	Water from / to
Cooling towers	2.66 (2.18-3.09)	3.05 (2.49-3.52)	2.12 (1.78-2.48)	muni / sewer
Hybrid	0.64 (0.47-0.87)	2.35 (1.93-2.88)	0.51 (0.37-0.70)	muni / sewer
Closed loop with assist	0.07 (0.04-0.11)	0.37 (0.23-0.55)	0.05 (0.03-0.09)	muni / sewer
Closed loop	<0.01	<0.01	<0.01	muni / sewer

Checks on this snapshot

Coverage of the state's places: water system 95%, watersheds 100%, utilities 100%, climate division 100%, a sewer plant within 40 km 91%. 50 places chosen at random ran through every step and cooling design with 0 errors. Places with no mapped water system are mostly rural areas on private wells.

Limits

These are screening estimates from public data, for comparing designs and sizes before an applicant's numbers exist. They do not replace a facility's design data, a utility's capacity analysis or a permit review. The watershed path is followed watershed to watershed, not stream by stream. EPA assessments reflect each state's latest report. Weather is the past 30 years.

Planning-level hydrologic assessment. Not engineering design. Built with the open waterledger Python packages (waterledger 2.0.0); methods, parameters and sources: waterledger.org and the package repository. Northline Hydrology.